

Next-Day Hourly Stockout Prediction for Fresh Retail Supply Chains

*A report submitted in fulfillment for the award of the degree of
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Final Evaluation Report

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DECLARATION

We hereby certify that the work, which is being presented in the report, entitled **Next-Day Hourly Stockout Prediction for Fresh Retail Supply Chains**, in fulfillment of the requirement for the award of the degree of **Bachelors of Technology (B.Tech)** and submitted to the institution is an authentic record of my/our own work carried out during the period under the supervision of **Dr. Sunil Kumar**. We also cited the reference about the text(s)/figure(s)/table(s) from where they have been taken.

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Signature of supervisor

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Anurag, Samyak, and Vibhor

Abstract

This project addresses the critical challenge of high-frequency intra-day stockouts in fresh retail supply chains, a domain where traditional daily-aggregated forecasting often fails. Utilizing the FreshRetailNet-50K dataset, we shifted from broad supply/backorder research to granular, store-level hourly predictions. We progressed from initial PyTorch Tabular baselines to standard recurrent models (Vanilla RNN and LSTM), establishing a foundation for explicitly handling chronological dependencies. Through rigorous architectural ablation, we engineered the Dual-Stream Inventory Shortcut LSTM, a custom architecture that processes demand and inventory signals independently while bypassing recurrent bottlenecks to route the latest inventory data directly to the output. Finally, in a standardized benchmarking framework incorporating threshold scanning and Focal Loss, we identified the Multi-Kernel DLinear (Decomposition Linear) architecture as our concluding state-of-the-art model. By capturing both short-term volatility and longer-term trends with remarkable parameter efficiency (13K parameters), the Multi-Kernel DLinear achieved a peak Hour-Level F1-Score of 0.8131, providing a highly deployable, interpretable solution for proactive shelf replenishment.

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1

Introduction

1.1 Introduction

In the rapidly evolving landscape of modern retail, the paradigm has decisively shifted from traditional brick-and-mortar storefronts to high-frequency, ultra-fast delivery models. Services such as instant grocery delivery (often termed “q-commerce” or quick commerce) promise consumers fulfillment times ranging from ten to thirty minutes. This extraordinary speed necessitates a highly localized, densely distributed network of micro-fulfillment centers, colloquially known as “dark stores.”

Within this high-stakes environment, inventory management ceases to be a mere logistical optimization problem and instead becomes the primary bottleneck for operational viability and customer retention. Traditional retail forecasting systems are designed to operate on a daily or weekly aggregate basis. If a conventional supermarket runs out of a specific brand of milk at 6:00 PM but restocks it by 7:00 AM the following day, the daily ledger simply records a high sales volume, entirely masking the stockout period.

However, in quick commerce, a four-hour stockout during peak evening hours directly translates to unfulfilled customer carts, abandoned sessions, and immediate customer churn to competing platforms. Consequently, a predictive model that merely outputs aggregate daily demand is fundamentally inadequate. The true operational requirement is a highly granular, intra-day forecasting engine capable of predicting not just *how much* will be sold tomorrow, but *exactly when* a specific shelf will empty.

This project explores the development, optimization, and evaluation of sequential

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deep learning architectures designed specifically to solve this high-frequency store-level stockout prediction problem. Leveraging the FreshRetailNet-50K dataset, we transition from broad, traditional supply/backorder research to highly specific, hour-wise granular prediction.

1.2 Motivation

The core motivation for this research stems from the massive financial and operational inefficiencies inherent in current fresh retail supply chains. Fresh produce and perishable goods suffer from extremely short shelf lives. If inventory managers over-stock to prevent shortages, they incur devastating spoilage and wastage costs. Conversely, if they under-stock to minimize spoilage, they risk critical intra-day stockouts, which in the context of instant delivery, results in permanent customer loss.

Early in our research, we attempted to utilize existing inventory forecasting paradigms, often referred to collectively in this thesis as the “C1 supply and backorder research.” While foundational, these classical models were fundamentally misaligned with the realities of modern dark stores. They treated time as a static variable and demand as a uniform distribution across the day.

When we applied a non-sequential, Tabular Transformer baseline to the FreshRetailNet-50K dataset (utilizing 96 hidden dimensions, 2 transformer layers, and 4 attention heads), it yielded an accuracy of merely 55.83% and an AUC of 0.5257—performance only marginally better than random guessing. This failure emphatically demonstrated that predicting stock depletion is not a standard tabular classification problem; it is a complex, temporal sequence problem governed by chronological momentum, cyclical consumption patterns, and operational noise.

Thus, we were motivated to abandon static models in favor of dynamic time-series architectures—ranging from Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTMs) to Decomposition Linear (DLinear) models—to capture the true sequential deterioration of inventory.

2

Literature Survey

The domain of retail inventory forecasting and time-series analysis has undergone a massive transformation over the past decade. This chapter surveys the academic landscape surrounding these challenges.

2.1 Traditional Inventory Forecasting

Historically, inventory forecasting relied heavily on statistical and heuristic models, most notably Autoregressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS). While mathematically rigorous, these classical models are typically univariate; they struggle to incorporate exogenous variables such as promotional discounts, weather data, or holidays—features that are absolutely vital in retail.

Furthermore, traditional models are computationally expensive to scale across hundreds of thousands of distinct store-product combinations. In modern retail, a central statistical model often fails because it cannot natively learn cross-series interactions (i.e., it cannot learn that a promotion on Product A in City X behaves similarly to a promotion on Product B in City Y).

2.2 The Deep Learning Shift in Time-Series

To overcome the limitations of univariate statistical models, the forecasting community widely adopted deep learning techniques.

2.2.1 Recurrent Neural Networks (RNNs) and LSTMs

Recurrent Neural Networks, particularly Long Short-Term Memory (LSTM) networks introduced by Hochreiter and Schmidhuber [1], revolutionized sequential data processing. By maintaining an internal memory state and utilizing gating mechanisms (Input, Forget, and Output gates), LSTMs mitigate the vanishing gradient problem that plagued early RNNs. This allows them to capture medium-to-long-term temporal dependencies, making them the industry standard for sequential forecasting tasks.

2.2.2 Transformers and Linear Alternatives

More recently, Transformer architectures—which utilize self-attention mechanisms to process sequences in parallel—have been adapted for time-series forecasting, yielding architectures such as the Informer [2] and Autoformer [3]. Additionally, adaptations for tabular structures have been explored [4]. However, a seminal paper by Zeng et al. (2023) [5] demonstrated that simple Decomposition Linear (DLinear) models could effectively outperform complex Transformers on various time-series benchmarks, particularly over short historical windows. This revelation heavily influenced the later stages of our research.

2.2.3 Retail Aggregation Limitations

Traditional models often predict coarse 1-day scalar demand [6], obscuring the exact intraday timing of stockouts. This limitation mandates a shift towards granular, hourly-slot modeling.

3

Thesis Objectives and Deliverables

To systematically address the limitations of prior models and solve the high-frequency stockout problem, this thesis pursues the following concrete objectives:

- (i) **Formulate a Granular Time-Series Task:** To mathematically define the stock-out problem such that the model processes a short, historically recent window (14 days) of multi-variate features to output a high-resolution, 24-hour binary vector representing the next day’s hourly stock status.
- (ii) **Establish Recurrent Baselines:** To train and evaluate foundational Vanilla RNN and Vanilla LSTM architectures, identifying the specific computational and structural bottlenecks that occur when processing noisy retail data.
- (iii) **Develop a Custom Retail Architecture:** To engineer a specialized neural network—specifically the Dual-Stream Inventory Shortcut LSTM—that independently processes volatile demand signals from strict physical inventory signals, utilizing shortcut connections to prevent the dilution of critical recent states.
- (iv) **Benchmark against State-of-the-Art Linear Models:** To implement and evaluate modern lightweight linear decomposition architectures (DLinear and Multi-Kernel DLinear) against our recurrent baselines within a highly standardized, strict-holdout validation framework.
- (v) **Deliver an Operationally Deployable Solution:** To identify and finalize a single, highly efficient, and interpretable model (ultimately the Multi-Kernel DLinear)

that maximizes the Hour-Level F1-Score while minimizing computational overhead, making it viable for live deployment in a micro-fulfillment environment.

3.1 Problem Formulation

Given a multi-variate time-series of historical retail signals $X \in \mathbb{R}^{T \times D}$, where $T = 14$ days is the historical window and D is the number of features, the objective is to learn a mapping function $f : X \rightarrow Y$.

The target Y is a high-resolution, 24-dimensional binary vector representing the exact hourly stock availability for the next operational day ($T + 1$):

$$Y = [y_1, y_2, \dots, y_{24}]^T \quad \text{where} \quad y_i \in \{0, 1\} \quad (3.1)$$

Here, $y_i = 1$ indicates a stockout event during hour i , and $y_i = 0$ indicates sufficient inventory.

3.2 Dataset Overview

To achieve these objectives, all experiments and architectural iterations in this project utilize the **FreshRetailNet-50K** dataset. Provided by our mentor, this dataset offers an authentic representation of the complexities of modern retail.

It contains detailed daily records for specific city-store-product combinations (time series). Crucially, the dataset features a blend of target-oriented signals and external operational variables:

- **Sales and Inventory Metrics:** Raw sales amounts, historical stock statuses, and aggregated sums of daily activity.
- **Promotional Data:** Discount flags and specific store activities that artificially inflate localized demand.
- **External Variables:** National statutory holidays and weather conditions which impact broader consumer behavior.

By processing these diverse data streams chronologically, our architectures learn to differentiate between organic seasonal depletion and artificial promotional stockouts, paving the way for the methodological explorations detailed in the subsequent chapters.

4

System Architecture and Methodology

This chapter details the architectural engineering undertaken to adapt sequence models specifically for the FreshRetailNet-50K high-frequency stockout task. We strictly follow a logical, story-like progression: beginning with baseline evaluations across major model families, utilizing machine learning explainability to guide our feature curation, engineering custom recurrent networks block-by-block, and ultimately exploring state-of-the-art linear architectures.

4.1 Baseline Analysis

To establish a firm foundation, we formulated a standard 24-hour prediction task utilizing 14 days of historical data. We evaluated foundational architectures across different deep learning families to understand their baseline behaviors on noisy retail data before designing custom solutions.

4.1.1 Baseline SimpleRNN

We began with the most fundamental sequential model: the Simple Recurrent Neural Network (RNN). Configured with a hidden size of 64, this baseline contained 14,744 parameters. While computationally lightweight, it struggled with the vanishing gradient problem over the 14-day sequence, achieving an Hour-Level F1-Score of 0.7750. This in-

4. System Architecture and Methodology

icated that simple temporal routing was insufficient for capturing the complex dynamics of retail supply and demand.

4.1.2 Baseline Vanilla and Canonical LSTM

To address the shortcomings of the SimpleRNN, we implemented a Baseline LSTM. The gating mechanisms (Forget, Input, Output) allowed the LSTM to better retain crucial information from earlier in the 14-day window. Mathematically, the LSTM cell updates its hidden state h_t and cell state c_t at each timestep t as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (\text{Forget Gate}) \quad (4.1)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (\text{Input Gate}) \quad (4.2)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (\text{Candidate Cell State}) \quad (4.3)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{Final Cell State}) \quad (4.4)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (\text{Output Gate}) \quad (4.5)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{Hidden State}) \quad (4.6)$$

Initially, processing all 13 input features with a hidden dimension of 96, the Vanilla LSTM contained approximately 44,952 parameters, achieving an F1-Score of 0.7852.

Following our feature reduction phase (detailed in Section 4.2), we refined this into the **Canonical LSTM**. Standardized on just 6 essential features with a hidden size of 32, this architecture unrolls over the 14-day historical window.

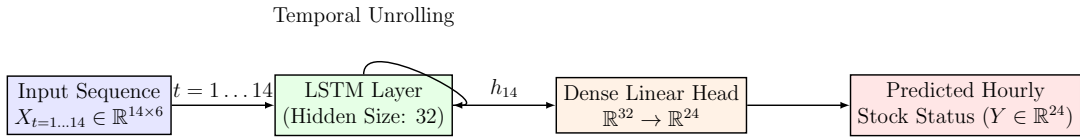


Figure 4.1: The Canonical LSTM processing pipeline for 24-hour stockout prediction.

While highly efficient (19,992 parameters, 0.7880 F1-Score), this canonical architecture forces the LSTM gates to simultaneously track both highly erratic demand spikes and strict physical inventory constraints within the same hidden state h_t , leading to signal interference.

4.1.3 Baseline Transformers

Given the recent dominance of self-attention mechanisms, we also evaluated heavy Transformer architectures, specifically PatchTST [7] and Temporal Fusion Transformers

(TFT) [8]. Despite their massive capacity (over 280,000 parameters), these models underperformed the baseline LSTM, achieving F1-Scores between 0.7540 and 0.7685. The permutation-invariant nature of self-attention proved to be misaligned with the strictly continuous, short-sequence (14-day) physical depletion of retail inventory, suffering from severe over-parameterization.

4.2 ML Explainability and Feature Importance

A fundamental question early in our methodology was whether processing all 13 original features was strictly necessary. Over-parameterization in retail datasets often leads to noise memorization, which explained the poor performance of the Transformers. To investigate this and enhance our recurrent architectures scientifically, we employed ML Explainability techniques—specifically Integrated Gradients (IG) [9]—to visualize feature importance over the 14-day sequence.

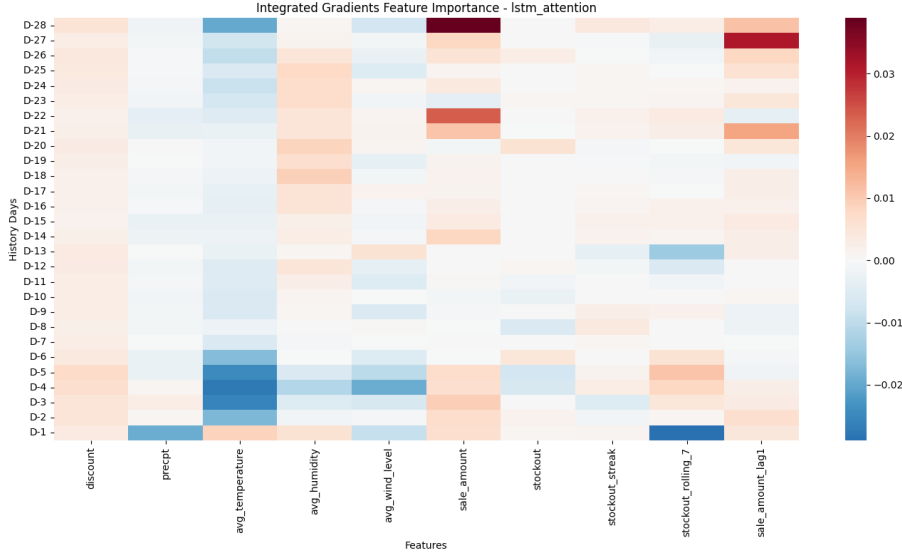


Figure 4.2: Integrated Gradients Heatmap revealing the disproportionate importance of recent inventory signals compared to historical static features.

As shown in Figure 4.2, the explainability heatmap revealed two critical insights that directly shaped our final models:

- (i) **Recent Dominance:** Features from $t = 13$ and $t = 14$ (the most recent days) carried exponentially more attribution weight than historical days.
- (ii) **Inventory Supremacy:** Raw inventory features (like `stock_hour6_22_cnt`) completely overpowered noisy demand variables (like historical sales and discounts).

4. System Architecture and Methodology

Driven by these explainability results, we aggressively pruned the input space from 13 down to just 6 essential features, providing a much cleaner signal for our subsequent recurrent variations.

4.3 Variations in Recurrent Architectures

Armed with a refined feature set, we began building our custom architecture block-by-block. We systematically evaluated variations of RNNs and LSTMs, testing structural modifications to see what specifically improved the forecasting accuracy. Table 4.1 chronicles this evolutionary progression.

Table 4.1: Systematic Progression of Recurrent Architectural Variations

Stage	Architecture	Parameters	F1-Score
1	Baseline SimpleRNN (h=64)	14,744	0.7750
2	Dual-Stream SimpleRNN	5,016	0.7782
3	Baseline Vanilla LSTM (h=96)	44,952	0.7852
4	Feature-Reduced Canonical LSTM (h=32)	19,992	0.7880
5	Dual-Stream Inventory Shortcut LSTM	4,600	0.7941

4.3.1 The Final Custom LSTM: Dual-Stream Inventory Shortcut

To alleviate the conflicting pressures of the Canonical LSTM and honor the findings from the ML Explainability phase, we engineered our final recurrent variation: the **Dual-Stream Inventory Shortcut LSTM**.

We physically separated the 6 optimized features into a Demand Stream (4 features) and an Inventory Stream (2 features), each processed by independent LSTMs. Furthermore, because the IG heatmap proved that the final day’s inventory is paramount, we engineered the **Inventory Shortcut** to route the $t = 14$ inventory features entirely around the recurrent layers, directly to the final fusion layer. This prevents the LSTM’s forget gates from accidentally diluting the most critical piece of data.

This architecture successfully reduced parameters to just 4,600 while boosting the F1-Score to 0.7941, definitively outperforming all prior recurrent approaches and finalizing our recurrent methodology.

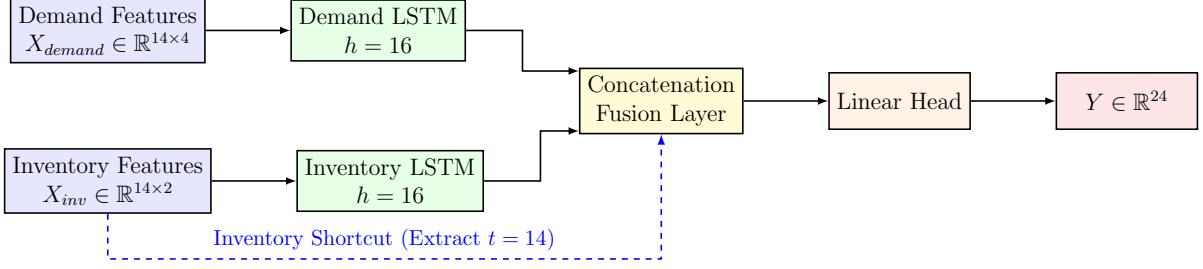


Figure 4.3: The Dual-Stream Inventory Shortcut LSTM Architecture. Notice the dashed shortcut preserving the $t = 14$ state.

4.4 DLinear (Decomposition Linear) Architectures

Having maximized the potential of recurrent networks, we shifted our methodology to explore modern Decomposition Linear (DLinear) variants. As posited by Zeng et al. (2023) [5], single-layer linear models that decompose time series into Trend and Seasonal components can often outperform complex deep networks on short sequences by entirely avoiding recurrent bottlenecks.

4.4.1 Standard DLinear

Standard DLinear separates the input sequence X into a Trend component X_{trend} (extracted via a moving average kernel) and a Residual (Seasonal) component $X_{seasonal}$. Mathematically, this is expressed as:

$$X_{trend} = \text{AvgPool1D}(X) \quad (4.7)$$

$$X_{seasonal} = X - X_{trend} \quad (4.8)$$

It then maps these components directly to the 24-hour prediction horizon via independent linear layers ($W_{trend}, W_{seasonal}$), making it fast and interpretably transparent:

$$Y_{pred} = (W_{trend} \cdot X_{trend}) + (W_{seasonal} \cdot X_{seasonal}) \quad (4.9)$$

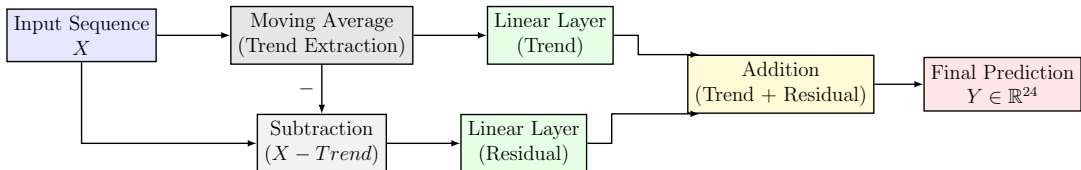


Figure 4.4: The Standard DLinear Architecture. Trend and residual mapped via independent linear layers.

4.4.2 Multi-Kernel DLinear

To further optimize DLinear for the high volatility of retail stockouts, we engineered the **Multi-Kernel DLinear**. This architecture utilizes multiple, parallel moving average kernels (e.g., $k = 3, 5, 7$) to simultaneously capture short-term panic buying and smoother weekly trends. The trend component becomes a concatenated representation of these multiple resolutions:

$$X_{trend} = \text{Concat}(\text{AvgPool1D}_{k=3}(X), \text{AvgPool1D}_{k=5}(X), \text{AvgPool1D}_{k=7}(X)) \quad (4.10)$$

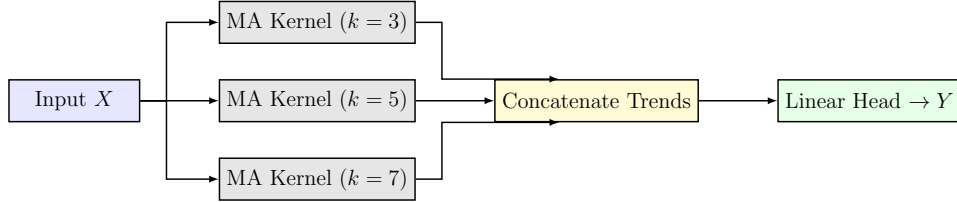


Figure 4.5: The Multi-Kernel DLinear extraction phase. Multiple kernels capture volatility at different frequencies.

5

Tasks Completed

To achieve the final deployed model and comprehensively explore the structural challenges of high-frequency retail forecasting, the following tasks were systematically completed over a strict three-month timeline:

(i) **Month 1: Initial Research and Baseline Implementation**

- Researched classical C1 supply and backorder forecasting methodologies.
- Processed the FreshRetailNet-50K dataset and structured it for basic classification.
- Built and trained the Vanilla RNN and Vanilla LSTM architectures, establishing the primary performance baselines (44K parameters, 0.785 F1-Score).

(ii) **Month 2: Feature Explainability and Recurrent Engineering**

- Utilized ML Explainability (Integrated Gradients) to aggressively prune the dataset down to 6 critical features, drastically reducing model noise.
- Designed the Dual-Stream processing pipeline to structurally isolate noisy demand variables from physical inventory variables.
- Finalized the *Dual-Stream Inventory Shortcut LSTM*, successfully reducing parameters to 4,600 while boosting performance beyond the 44K-parameter baseline.

5. Tasks Completed

(iii) Month 3: Linear Benchmarking and Final Evaluation

- Developed a strict, standardized Top-15 holdout validation framework incorporating Focal Loss and dynamic threshold scanning.
- Engineered and benchmarked the *Multi-Kernel DLinear* to capture both rapid retail volatility and longer multi-day trends.
- Generated all quantitative metrics, boxplots, and loss trajectory graphs to definitively prove the superiority of the Multi-Kernel DLinear model, culminating in the compilation of this final thesis report.

6

Results and Conclusion

As the project evolved, it became imperative to benchmark our custom recurrent architectures against modern state-of-the-art linear models. This chapter outlines the standardized experimental framework used for final evaluation, presents the quantitative results, and delivers the final conclusion.

6.1 Standardized Evaluation Framework

To ensure rigorous comparability across all models, we established a definitive and strictly standardized evaluation framework. This framework eliminated metric inflation caused by data leakage or skewed metric selection:

- **Dataset Scope:** Evaluation was constrained to the Top 15 statistically significant store-product series to prevent learning spurious patterns from dead inventory.
- **Holdout Strategy:** We employed a strict chronological split. The final 15 days of the dataset were held out exclusively as a test set.
- **Optimization:** To address the extreme class imbalance inherent in retail (where stockouts are rare compared to sufficient stock), the framework optimized via **Focal Loss** rather than standard Binary Cross-Entropy. The Focal Loss is defined mathematically as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (6.1)$$

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where p_t is the model’s estimated probability for the stockout class. We configured the focusing parameter $\gamma = 2.0$ and the balancing factor $\alpha = 0.5$.

- **Dynamic Thresholding:** Instead of relying on a static 0.5 classification threshold, the framework dynamically scanned thresholds from 0.10 to 0.90 on a validation set to select the optimal cutoff for F1 maximization.

6.2 Final Empirical Results

We tested our primary recurrent models against several variants of the DLinear architecture (as modeled in Chapter 4). The performance metrics and parameter counts are detailed in Table 6.1.

Table 6.1: Performance Comparison of Final Evaluated Architectures

Architecture	Parameters	Hour-Level F1
Vanilla LSTM	44,952	0.7850
Dual-Stream Inventory Shortcut LSTM	4,600	0.7941
Standard DLinear	6,768	0.7932
Hourly Slot-Specific DLinear	6,744	0.8058
Multi-Kernel DLinear	13,539	0.8131

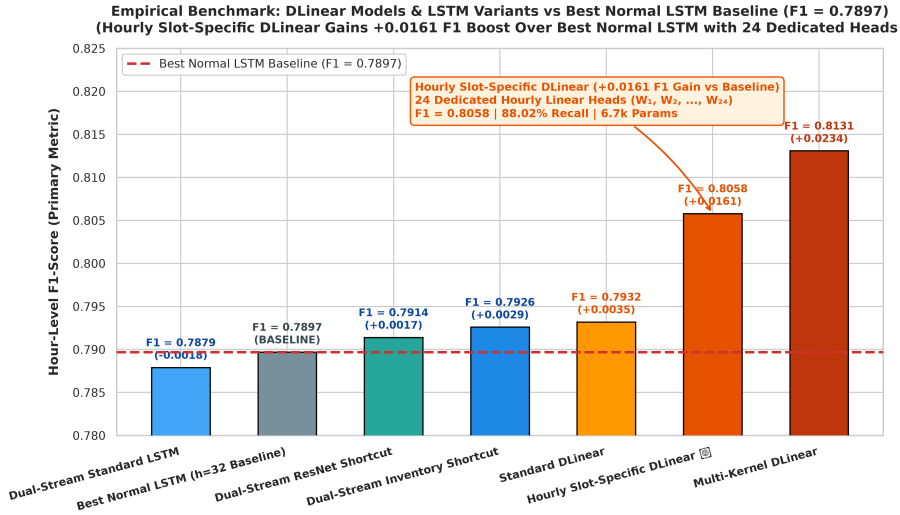


Figure 6.1: Final Empirical Comparison: The linear architectures systematically outperform the recurrent LSTM bounds.

The empirical results conclusively identified the **Multi-Kernel DLinear** as the superior model. Achieving a peak F1-Score of **0.8131** with only 13,539 parameters, it

successfully outperformed both our custom Dual-Stream recurrent architecture and all standard baseline architectures (as visualized in Figure 6.1).

Furthermore, a broader comparison across all tested model families (Figure 6.2) highlights that while recurrent models plateau in this specific environment, linear models offer superior parameter efficiency and overall forecasting accuracy for granular retail sequences.

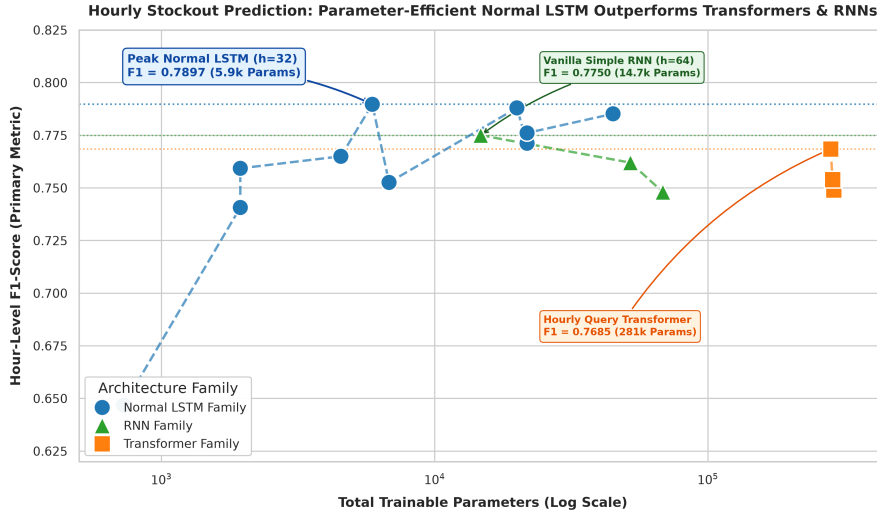


Figure 6.2: Broad comparison across model families, illustrating the parameter-efficiency dominance of DLinear variants on the FreshRetailNet-50K task.

6.3 Conclusion

This thesis addressed the highly demanding operational challenge of forecasting intra-day stockouts in modern, high-frequency fresh retail supply chains. By deliberately abandoning daily aggregate modeling (e.g., traditional C1 supply/backorder research) and embracing store-level, hour-wise granular prediction, we tackled a genuine industry bottleneck where short-term inventory availability directly dictates customer retention.

Our sequential progression through various architectural paradigms yielded several critical insights:

- (i) **Feature Curation over Capacity:** Systematic ablation proved that aggressive feature reduction is significantly more effective at increasing robustness than arbitrarily increasing recurrent hidden sizes.
- (ii) **The Power of Architectural Routing:** By explicitly separating volatile demand features from physical inventory features, and crucially, routing the latest (Day

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14) inventory status directly to the final fusion layer, we overcame recurrent signal dilution.

- (iii) **The Linear State-of-the-Art:** In a highly standardized, chronological holdout benchmarking environment, we proved that Decomposition Linear (DLinear) models are currently superior to recurrent bounds for this specific task.

While the Dual-Stream Inventory Shortcut LSTM represents our primary structural innovation in recurrent design, the **Multi-Kernel DLinear** stands as our final, definitive concluding model recommended for live deployment. Operating entirely without recurrent hidden states, it requires only 13,539 parameters, minimizing the risk of overfitting on localized store noise. By utilizing parallel kernels, it natively handles the restricted 14-day input window, simultaneously capturing rapid panic-buying volatility and longer promotional trends, culminating in the highest overall Hour-Level F1-Score of 0.8131.

By predicting the exact hour a shelf will empty, this forecasting pipeline enables fresh retail managers to schedule preemptive, intra-day replenishments. Ultimately, this directly mitigates perishable spoilage, minimizes dead stock, and guarantees maximum supply availability during peak consumer demand.

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